

Sweet Bread Production Process Analysis – PT XYZ, Yogyakarta

Factory Name: PT XYZ (nama disamarkan/pseudonim sesuai publikasi asli) — produsen roti manis berskala kecil-menengah di Yogyakarta, Indonesia.

Process Type: Serial (sequential flow shop). Every unit passes through all 10 stages sequentially on a single production line, with batch-level pipelining between stages.

Factory Description: PT XYZ is a small-to-medium scale bakery producing sweet bread as its main product. The line runs from raw material weighing to finished goods packaging, with four quality control points at Mixing, Filling & Panning, Proofing, and Baking, plus one manual Sorting stage before packaging.

Number of Workers: ±14 workers per production shift.

TABLE 1 — Process Stages, Equipment, Tasks, Automation, QC, Throughput

Process Stage	Equipment	Operator Task Description	Automatio n	Quality Control (QC)	Throughput (estimated)
Weighing	Digital Weighing Scale	Menimbang bahan sesuai formulasi.	✓	—	±2.500–3.000 pcs-eq/jam (mendukung ±5–6 batch adonan 25 kg/jam). Bukan bottleneck.
Mixing	Mixer	Memasukkan bahan, mengoperasikan, dan memantau proses.	✓	Memastikan adonan homogen, elastis, dan tidak lengket.	±1.500 pcs/jam (±3 batch adonan @25 kg, ±18–20 menit/batch; 1 pcs ±50 g).
Dough Dividing	Dough Divider	Mengoperasikan dan memantau proses.	✓	—	±1.800–2.000 pcs/jam (kapasitas tipikal dough divider semi-otomatis skala menengah).
Dough Shaping	Rounder Machine / Dough Sheeter	Mengoperasikan dan memantau proses.	✓	—	±1.800–2.000 pcs/jam, sejalan dengan laju keluaran Dough Divider.
Filling & Panning	Breadline Machine	Menata adonan pada baking tin dan memantau proses.	✓	Memastikan jarak antar adonan ±4 cm.	±1.200–1.400 pcs/jam — dibatasi kecepatan pengisian/penataan manual oleh operator.
Proofing	Proofing Cabinet	Mengatur suhu dan waktu proofing.	✓	Memastikan suhu 55–60°C selama 50–60 menit serta adonan mengembang dengan baik (volume roti).	±1.300–1.500 pcs/jam — proses batch, siklus 50–60 menit per rak/trolley.
Baking	Deck Oven / Combi Oven	Memasukkan dan mengeluarkan loyang serta mengatur parameter pemanggangan.	✓	Memastikan suhu 170°C selama 8–10 menit sehingga kematangan roti sesuai dan tidak gosong (warna roti).	±700–750 pcs/jam (1 unit oven) — bottleneck lini produksi; siklus bakar 8–10 menit/loyang.
Cooling	Cooling Area	Memindahkan produk untuk proses pendinginan.	—	—	±700–750 pcs/jam — mengikuti laju keluaran Baking, area terbuka tanpa batas kapasitas ketat.
Sorting	—	Memisahkan produk cacat dan produk yang memenuhi standar.	—	Memeriksa produk cacat: gosong, filling bocor, lengket, hancur, kurang matang, tidak mengembang, tanpa topping, ukuran tidak sesuai, dan cacat lainnya.	±800–900 pcs/jam (2 operator QC manual) — melampaui laju Baking, bukan bottleneck.
Packaging	Packaging Machine	Mengemas produk yang telah lolos sorting.	—	—	±1.000–1.200 pcs/jam (mesin kemas otomatis) — bukan bottleneck.

TABLE 2 — Worker Allocation per Process Stage

Process Stage	Number of Workers	Assigned Workers	Assignment Notes
Weighing	1 orang	Pekerja 1	Menimbang seluruh bahan formulasi sebelum mixing.
Mixing	1 orang	Pekerja 2	Mengoperasikan mixer & memantau kekalisan adonan.
Dough Dividing	1 orang	Pekerja 3	Mengoperasikan dough divider.
Dough Shaping	1 orang	Pekerja 4	Mengoperasikan rounder/sheeter.
Filling & Panning	2 orang	Pekerja 5, 6	Menata adonan ke loyang & mengatur jarak ±4 cm (kerja manual paralel).
Proofing	1 orang	Pekerja 7	Mengatur suhu/waktu proofing cabinet.
Baking	2 orang	Pekerja 8, 9	Memasukkan/mengeluarkan loyang & memantau parameter oven (kerja panas, bergiliran).
Cooling	1 orang	Pekerja 10	Memindahkan produk ke area pendinginan.
Sorting	2 orang	Pekerja 11, 12	QC manual memisahkan produk cacat vs. standar.
Packaging	2 orang	Pekerja 13, 14	Mengoperasikan mesin kemas & mengemas produk lolos sorting.

TABLE 3 — Equipment per Stage and Unit Count

Process Stage	Equipment	Number of Units	Capacity per Unit	Total Stage Capacity
Weighing	Digital Weighing Scale	1 unit	±2.500–3.000 pcs-eq/jam	±2.500–3.000 pcs-eq/jam
Mixing	Mixer (kapasitas ±25 kg/batch)	1 unit	±1.500 pcs/jam	±1.500 pcs/jam
Dough Dividing	Dough Divider	1 unit	±1.800–2.000 pcs/jam	±1.800–2.000 pcs/jam
Dough Shaping	Rounder Machine + Dough Sheeter	2 unit (1 rounder + 1 sheeter, dipakai sesuai varian produk)	±1.800–2.000 pcs/jam/unit	±1.800–2.000 pcs/jam
Filling & Panning	Breadline Machine + Loyang (baking tin)	1 unit breadline + ±50 unit loyang (sirkulasi Filling-Proofing-Baking-Cooling)	±1.200–1.400 pcs/jam (breadline)	±1.200–1.400 pcs/jam
Proofing	Proofing Cabinet	1 unit	±1.300–1.500 pcs/jam	±1.300–1.500 pcs/jam
Baking	Deck Oven + Combi Oven	2 unit (1 deck oven + 1 combi oven, paralel/cadangan)	±700–750 pcs/jam/unit	±1.400–1.500 pcs/jam gabungan
Cooling	Cooling Rack / Trolley	±10 unit (buffer ±15 menit dwell time)	—	Mengikuti laju Baking
Sorting	Meja Sortir + Keranjang Sortir	2 meja (sesuai 2 operator QC)	±400–450 pcs/jam/meja	±800–900 pcs/jam
Packaging	Packaging Machine	1 unit	±1.000–1.200 pcs/jam	±1.000–1.200 pcs/jam